String Functions

Time limit: 1.00 s **Memory limit:** 512 MB

We consider a string of n characters, indexed $1, 2, \dots, n$. Your task is to calculate all values of the following functions:

- $z(i)$ denotes the maximum length of a substring that begins at position i and is a prefix of the string. In addition, $z(1) = 0$.
- $\pi(i)$ denotes the maximum length of a substring that ends at position i, is a prefix of the string, and whose length is at most $i - 1$.

Note that the function z is used in the Z-algorithm, and the function π is used in the KMP algorithm.

Input

The only input line has a string of length n. Each character is between a–z.

Output

Print two lines: first the values of the z function, and then the values of the π function.

Constraints

- $1 \le n \le 10^6$

Example

Input	Output
`abaabca`	`0 0 1 2 0 0 1` `0 0 1 1 2 0 1`